

05th March '26 (Thu)

Assignment on estimates of quantities related to the electrical conductivity problem

Estimates of microscopic quantities

② Order of magnitude of $\langle \text{kinetic energy} \rangle$ of electron at room temperature is (in eV)
(a) 10^{-1} (b) 10^{-2} (c) 10^{-3} (d) 10^{-4}

$$\langle K \rangle = \frac{3}{2} k_B T \quad \text{--- (A)} \quad [IM]$$

@ $T = 300 \text{ K}$ and $= 3 \cdot 10^2 \text{ K}$, and

$$\parallel k_B = 1.38 \cdot 10^{-23} \text{ J/K},$$

$$\langle K \rangle = \frac{3}{2} \cdot 1.38 \cdot 10^{-23} \cdot 3 \cdot 10^2$$

$$\approx \frac{9}{2} \cdot 1.4 \cdot 10^{-21} \text{ J}$$

$$= \frac{12.6}{2} \cdot 10^{-21} \text{ J}$$

$$= 6.3 \cdot 10^{-21} \text{ J}$$

So,

$$\langle K \rangle (\text{in eV}) = \frac{6.3 \cdot 10^{-21} \text{ J}}{1.602 \cdot 10^{-19} \text{ J/eV}}$$

$$\parallel 1 \text{ eV} = 1.602 \cdot 10^{-19} \text{ J}$$

$$\approx 4 \cdot 10^{-21+19}$$

$$= 4 \cdot 10^{-2} \text{ eV}$$

$$\therefore \text{at } 300 \text{ K} \quad \langle K \rangle = 10^{-2} \text{ eV}$$

② Order of magnitude of thermal velocity of electron at room temperature is (in ms^{-1})
 (a) 10^4 (b) 10^5 (c) 10^6 (d) 10^2

$$\langle K \rangle = \frac{1}{2} m v_{th}^2 = \frac{3}{2} k_B T \quad \text{--- (1M) } \textcircled{A}$$

where v_{th} = thermal velocity

$$\begin{aligned} \textcircled{a} \quad T &= 300 \text{ K}, \quad \langle K \rangle = 4 \cdot 10^{-2} \text{ eV} \\ &= 4 \cdot 10^{-2} \cdot 1.602 \cdot 10^{-19} \text{ J} \\ &= 6.4 \cdot 10^{-21} \text{ J} \end{aligned}$$

So

$$\frac{1}{2} m v_{th}^2 = 6.4 \cdot 10^{-21} \text{ J} \quad // 1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$$

$$v_{th}^2 = \frac{2 \cdot 6.4 \cdot 10^{-21} \text{ kg m}^2 \text{ s}^{-2}}{9.1 \cdot 10^{-31} \text{ kg}}$$

$$// m_{\text{electron}} = 9.1 \cdot 10^{-31} \text{ kg}$$

$$v_{th}^2 = \frac{2 \cdot 6}{9.1} 10^{+10} \text{ m}^2 \text{ s}^{-2}$$

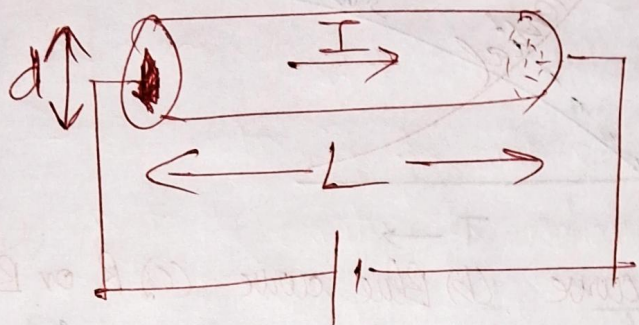
$$v_{th} = \sqrt{1.33} \cdot \sqrt{10^{+10}} \text{ m}^2 \text{ s}^{-2}$$

$$= 1.1 \cdot 10^{+5} \text{ ms}^{-1}$$

$$= 1.1 \cdot 10^{+5} \text{ ms}^{-1}$$

$$\therefore \textcircled{O} (v_{th}^{300\text{K}}) \approx 10^{+5} \text{ ms}^{-1}$$

④ Order of magnitude of electric field inside a copper wire (in Vm^{-1}) -



$$V = 0.5 \text{ V}$$

$$I = 22 \text{ A}$$

$$L = 2 \text{ m}$$

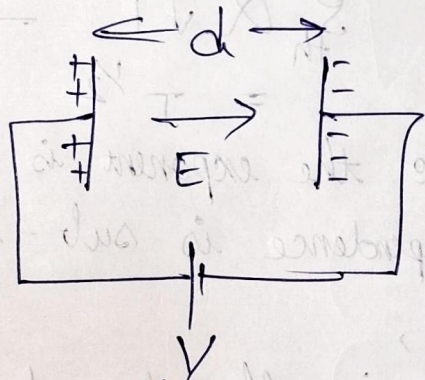
$$d = 1 \text{ mm}$$

(a) 10^0 (b) 10^1 (c) 10^{-1} (d) 10^{+2}

Ohm's law: $V = RI$

[M]

However, it is not needed for this problem.



$$E = \frac{V}{d} \quad \text{--- (A)}$$

$$= \frac{0.5 \text{ V}}{1 \text{ mm}}$$

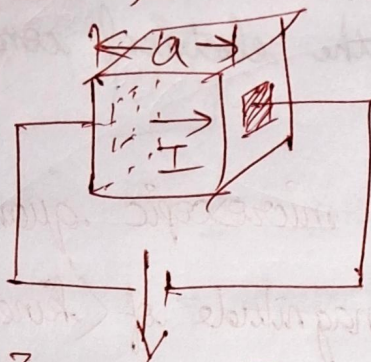
$$= \frac{0.5 \text{ V}}{10^{-3} \text{ m}}$$

$$= 5 \cdot 10^{-1} \cdot 10^{+3} \text{ Vm}^{-1}$$

$$= 5 \cdot 10^{+2} \text{ Vm}^{-1}$$

$$\therefore O(E) = 10^{+2} \text{ Vm}^{-1}$$

⑤ Order of magnitude of conductivity
(in Sm^{-1})



$$V = 100\text{V}$$

$$I = 1\text{nA}$$

$$a = 10\text{cm}$$

(a) 10^{-7} (b) 10^{+7} (c) 10^{-12} (d) 10^{-10}

Ohm's law: $V = RI$

$$\Rightarrow R = \frac{V}{I}$$

$$= \frac{100\text{V}}{1\text{nA}}$$

$$= \frac{10^2}{10^{-9}} \frac{\text{V}}{\text{A}}$$

$$= 10^{11} \Omega$$

Resistivity $\leftarrow$ Resistance

$$R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l} \quad \text{--- (A)}$$

$$= \frac{R \cdot a \cdot a}{a}$$

$$= Ra$$

$$= 10^{11} \cdot 10\text{cm}$$

$$= 10^{12} \cdot 10^{-2}\text{m}$$

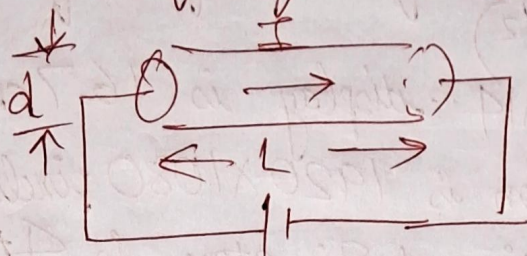
$$= 10^{10} \Omega\text{m}$$

$$\therefore \textcircled{C} (\sigma) = 10^{-10} \Omega^{-1}\text{m}^{-1}$$

Conductivity $\leftarrow$ Resistivity

$$\textcircled{B} \quad \sigma = \frac{1}{\rho} \Rightarrow \sigma = 10^{-10} \Omega^{-1}\text{m}^{-1}$$

⑥ Order of magnitude of drift velocity (in ms^{-1})



$$V = 0.5 \text{ V}$$

$$I = 22 \text{ A}$$

$$d = 1 \text{ mm}$$

$$L = 2 \text{ m}$$

$$n_{\text{Cu}} = 8.47 \cdot 10^{28} \text{ m}^{-3}$$

(a) 10^{-2}

(b) 10^{-3}

(c) 10^{+2}

(d) 10^{+3}

[2M]

Drift velocity $\leftarrow$ Current density

$$j = ne v_d \Rightarrow v_d = \frac{j}{ne}$$

$$\text{// } e = 1.602 \cdot 10^{-19} \text{ J}$$

(A)

Current density $\leftarrow$ Current

$$j = \frac{I}{A}$$

$$\Rightarrow j = \frac{I}{\frac{\pi d^2}{4}} = \frac{4I}{\pi d^2} \quad \text{--- (B)}$$

$$= \frac{4 \cdot 22 \cdot 10^{+1} \text{ A}}{3.14 (10^{-3})^2 \text{ m}^2}$$

$$\approx \frac{8.8}{3} \cdot 10^7 \text{ Am}^{-2}$$

$$j = 2.93 \cdot 10^7 \text{ Am}^{-2}$$

So, from (A)

$$v_d = \frac{2.93 \cdot 10^7}{8.47 \cdot 10^{+28} \cdot 1.602 \cdot 10^{-19}}$$

$$\approx \frac{3}{8.5 \cdot 1.6} \cdot 10^{+7-9} \text{ ms}^{-1}$$

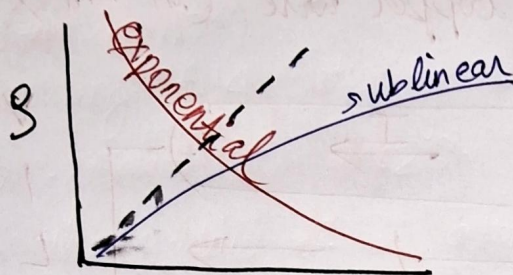
$$\therefore (v_d) =$$

$$10^{-3} \text{ ms}^{-1}$$

$$\approx \frac{3}{13.6} \cdot 10^{-2} = \frac{3}{14} \cdot 10^{-2} = 0.21 \cdot 10^{-2}$$

$$= 2.1 \cdot 10^{-3} \text{ ms}^{-1}$$

⑦ Metals behave as



(a) Red curve (b) Blue curve (c) R or ρ (d) None
Metals Classical Free Electron Theory (CFET) [3M]

predicts square root dependence on temperature of resistivity of metal i.e.

$$\rho_{th}^{metal} \propto \sqrt{T} \quad \text{--- (A)}$$

$$= T^{1/2}$$

Since the exponent is less than 1, the dependence is sub-linear whereas,

the experimentally observed dependence of resistivity on temperature is linear i.e.

$$\rho_{exp}^{metal} \propto T \quad \text{--- (B)}$$

$$= T^1$$

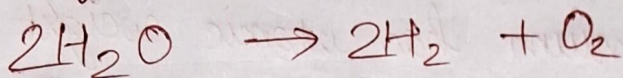
Semiconductors

On the other hand,

semiconductors have exponential form for ρ vs T i.e.

$\rho \propto \exp\left(\frac{1}{T}\right)$ or $\exp(-T)$?
 $\therefore$ Answer is linear (Find)

⑧ Order of magnitude of charge of a mole of electrons.



1 g of H_2 gas production requires how much charge (in C)?

(a) 10^{+3} (b) 10^{+4} (c) 10^{+5} (d) 10^{+6} [3M]

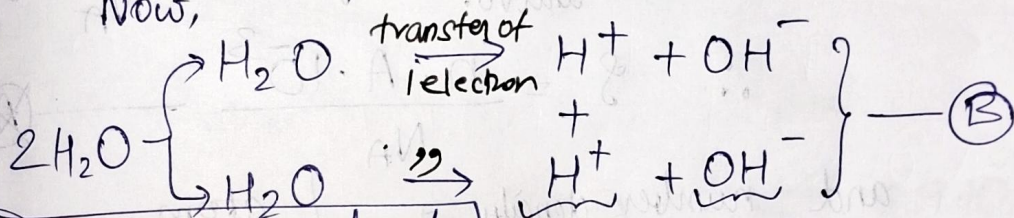
1 g of H_2 contains ? mol

1 mol H_2 weighs 2 g
 ? mol H_2 weighs 1 g — (A)

$$\Rightarrow ? \text{ mol} = 1 \text{ mol} \times \frac{1 \text{ g}}{2 \text{ g}} = 0.5 \text{ mol}$$

Now,

transfer of 1 electron



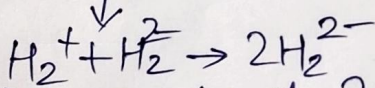
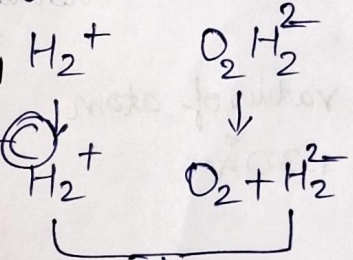
So charge transferred is $\frac{NA}{2} = 6.023 \cdot 10^{23} \text{ mol}^{-1}$

$$\Delta C = 0.5 NA \cdot e$$

$$= 3 \cdot 10^{23} \cdot 1.6 \cdot 10^{-19}$$

$$= 4.8 \cdot 10^{+4} \text{ C}$$

$$\Rightarrow \Delta C = 10^{+4} \text{ C}$$



So 2 molecules requires transfer of $2e^-$ to produce 2 molecules of H_2 gas
 $\Rightarrow$ 1 molecule of H_2 gas requires $1e^-$
 $\Rightarrow$ 1 mol of H_2 requires 1 mol e^-

$\Rightarrow$ 0.5 mol of H_2 requires $0.5 NA = 6.022 \cdot 0.5 \cdot 10^{23} e^-$ — (A)

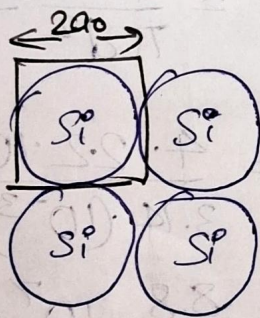
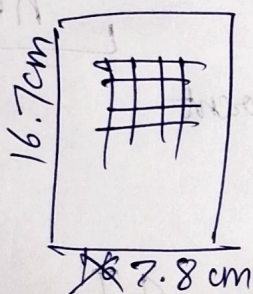
Q Order of magnitude of surface density
(~~in~~ in m^{-2})

A Samsung A14 display is $16.7 \text{ cm} \times 7.8 \text{ cm}$

Its resolution is 1920×1080 pixels.

If the radius of Si atom is $4 \text{ \AA}$, then the number of atoms that are seen in

one pixel of screen is of the order of
(a) 10^8 (b) 10^9 (c) 10^{10} (d) 10^{11} [3M]



Surface density of atoms (σ_{Si})

$$= \frac{\# \text{ of atoms on the surface}}{\text{surface area}}$$

$$= \frac{1 \text{ atom}}{\text{surface area of 1 atom}}$$

$$= \frac{\# \text{ of atoms in one pixel}}{\text{surface area of pixel}}$$

So, from (A),

$$\sigma_{\text{Si}} = \frac{1}{(2a_0)^2}$$

// a_0 = radius of Si atom

$$= \frac{1}{4 \cdot 16 \cdot (10^{-10})^2} \quad // 1 \text{ \AA} = 10^{-10} \text{ m}$$

$$= \frac{1}{6.4 \cdot 10^{-19}}$$

$$\approx \frac{10^{+19}}{6.4} = 1.6 \cdot 10^{+18} \text{ m}^{-2}$$

So, from (2) (B),

$$\sigma_{Si} = \frac{\# \text{ of atoms in one pinel}}{\text{area of pinel}} = 1.6 \cdot 10^{+18} \text{ m}^{-2}$$

Now area of pinel = $\left(\text{length of pinel} \right) \times \left(\text{breadth of pinel} \right)$

$$\# \text{ pinels}_{\text{length}} = 1920 = \left(\frac{\text{length of phone}}{\# \text{ pinels along length}} \right) \times \left(\frac{\text{breadth of phone}}{\# \text{ pinels along breadth}} \right)$$

$$\# \text{ pinels}_{\text{breadth}} = 1080$$

$$\text{length phone} = 16.7 \text{ cm}$$

$$\text{breadth phone} = 7.8 \text{ cm}$$

$$= \frac{16.7 \text{ cm}}{1920} \times \frac{7.8 \text{ cm}}{1080}$$

$$= \frac{17}{2000} \times \frac{8}{1000} \text{ cm}^2$$

$$= 6.9 \cdot 10^{+1-6} \cdot 10^{-4} \text{ m}^2$$

$$A_{\text{pinel}} = 6.9 \cdot 10^{-9} \text{ m}^2$$

$$\therefore \# \text{ atoms in 1 pinel} = \sigma_{Si} \cdot A_{\text{pinel}}$$

$$= 1.6 \cdot 10^{+18} \text{ m}^{-2} \cdot 6.9 \cdot 10^{-9} \text{ m}^2$$

$$= 1.6 \cdot 7 \cdot 10^{+9}$$

$$\# \text{ atoms} = 1.12 \cdot 10^{10}$$

$$\therefore \left(\frac{\text{atoms in a}}{\text{pinel}} \right) = 10^{10}$$

(16) Order of magnitude of number density (in m^{-3}).

Osmium has atomic weight of 190 and atomic number of 78. The radius of osmium is $1.30 \text{ \AA}$. The density of osmium metal is (in kg m^{-3}).

(a) 10^0 (b) 10^1 (c) 10^2 (d) 10^4 [3M]

(9) Mass density $\leftarrow$ Number density (n)

$Z_{\text{os}} = 78$ 1 atom weights A at weight

$A_{\text{os}} = 190$ 1 atom weighs $\frac{A}{N_A} \text{ g} = \frac{A \cdot 10^{-3}}{N_A} \text{ kg}$

(A) $\Rightarrow n \frac{\text{atoms}}{\text{unit vol}} \text{ weighs } n \cdot \frac{A \cdot 10^{-3}}{N_A} \frac{\text{kg}}{\text{unit vol}}$

$$\therefore \rho = \frac{n \cdot A \cdot 10^{-3}}{N_A} \quad \text{--- (A) (B)}$$

and number density = $\frac{1 \text{ atom}}{\text{vol of 1 atom}} \quad \text{--- (C)}$

$r = \text{radius of atom}$

$$= 1.30 \text{ \AA}$$

$$= \frac{1}{\frac{4}{3} \pi r^3}$$

$$= \frac{1}{\frac{4}{3} \cdot 3.14 \cdot (1.3 \cdot 10^{-10})^3}$$

$$= \frac{1}{4 \cdot 1.69 \cdot 1.3 \cdot 10^{-30}}$$

$$\approx \frac{1}{8} 10^{30}$$

$$n \approx 1.25 \cdot 10^{29} \text{ m}^{-3}$$

$$\begin{array}{r} 1.7 \\ \times 1.3 \\ \hline 2.01 \end{array}$$

So, from (A) & (B),

$$\rho = \frac{1.25 \cdot 10^{29} \cdot 1.90 \cdot 10^2 \cdot 10^{-3}}{6.023 \cdot 10^{23}}$$

$$\approx \frac{1.25 \cdot 2}{6} 10^5$$

$$= \frac{2.5}{6} 10^5$$

$$\rho_{os} = 4.16 \cdot 10^4 \text{ kg m}^{-3}$$

$$\Rightarrow \textcircled{C} (\rho_{os}) = 10^4 \text{ kg m}^{-3}.$$

① Order of magnitude of relaxation time
drift velocity (in ms^{-1}) $\xrightarrow{E}$

$$\tau_{cu} = 10 \text{ fs}$$

$$E = 10^3 \text{ Vm}^{-1}$$

What is v_d ?

- (a) 10^1 (b) 10^{-2} (c) 10^{+1} (d) 10^{+2}

[1M]

Drift velocity $\leftarrow$ Electric field

$$v_d = \mu E \longrightarrow \textcircled{A}$$

$$\text{where } \mu = \frac{e\tau}{m}$$

$$v_d = \frac{e\tau E}{m}$$

$$= \frac{1.602 \cdot 10^{-19} \cdot 10^{-14} \cdot 10^{+3}}{9.1 \cdot 10^{-31}}$$

$$= \frac{1.602}{9.1} 10^{+1}$$

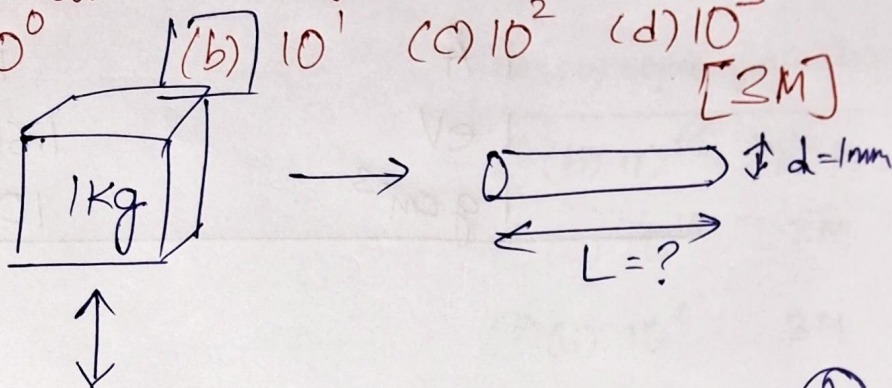
$$= 1.6 \cdot 10^0 \text{ ms}^{-1}$$

$$\therefore \textcircled{D} (v_d) = 10^0 \text{ ms}^{-1}$$

Since no option, free marks!

- ⑫ Order of magnitude of resistance in Ω
 $\sigma_{Cu} = 10^7 \text{ Sm}^{-1}$. A 1kg block of Cu
 is molded into a wire of diameter 1mm,
 If $\rho_{Cu} = 8.96 \text{ gcc}^{-1}$, what is R?
 (a) 10^0 (b) 10^1 (c) 10^2 (d) 10^3

[3M]



Vol of block ?

$$\text{mass density} = \frac{\text{mass of 1kg}}{\text{volume of 1kg}} = 8.96 \text{ g cm}^{-3} \quad \text{--- (A)}$$

$$\Rightarrow \text{Volume of 1kg} = \frac{1 \text{ kg}}{8.96 \cdot 10^3 \text{ kg m}^{-3}} = \frac{1}{8.96} \cdot 10^{-3} \text{ m}^3$$

$$V = 1.1 \cdot 10^{-4} \text{ m}^3$$

Now length of wire (L) $\leftarrow$ Volume of wire (V)

$$V = \left(\text{Area of cross section} \right) \times \text{Length} \quad \text{--- (B)}$$

$$V = \frac{\pi d^2}{4} \cdot L$$

$$\Rightarrow L = \frac{4V}{\pi d^2} = \frac{4 \cdot 1.1 \cdot 10^{-4} \text{ m}^3}{3.14 \cdot 10^{-6} \text{ m}^2}$$

$$\approx \frac{4.4}{3} \cdot 10^2 \text{ m}$$

$$L = 1.46 \cdot 10^2 \text{ m}$$

∴ Resistance ← Resistivity ← Conductivity

$$R = \frac{\rho L}{A} ; \quad \sigma = \frac{1}{\rho}$$

$$\Rightarrow R = \frac{L}{\sigma A} = \frac{4L}{\sigma \pi d^2}$$

$$= \frac{4 \cdot 46 \cdot 10^2 \text{ m}}{10^7 \Omega^{-1} \text{ m}^{-1} \cdot 10^{-6} \text{ m}^2} \quad \text{--- (C)}$$

$$= 4.15 \cdot 10^1 \Omega$$

$$= 6 \cdot 10^1 \Omega$$

$$\therefore \mathcal{O}(R) = 10^1 \Omega$$

⑬ Behaviour of metals

Look at ④⑦.

— End of assignment —

Marking scheme

Microscopic quantities		Macroscopic quantities	
② (b) 10^{-2}	1M	⑧ (b) 10^{+4}	*2.3M
③ (b) 10^{+5}	1M	⑨ (c) 10^{10}	3M
④ (d) 10^{+2}	1M	⑩ (d) 10^4	3M
⑤ (d) 10^{-10}	2M	⑪ Wrong question	1M
⑥ (b) 10^{-3}	2M	⑫ (b) 10^1	3M
⑦ (d) None	3M	⑬ (d) None	3M
10M		16M*	

*** So the brave students are having an opportunity to score more marks.

* The marking of a question is proportional to the number of ideas (or equations) required to solve the question.

Fundamental constants

Name	Symbol	Value
Boltzmann constant	k_B	$1.38 \cdot 10^{-23} \text{ J K}^{-1}$
Electron mass	m_e	$9.1 \cdot 10^{-31} \text{ kg}$
Electron charge	e	$1.602 \cdot 10^{-19} \text{ C}$
Avogadro constant	N_A	$6.023 \cdot 10^{23} \text{ mol}^{-1}$

Useful conversion factors

Physical quantity

In conventional units

In SI units

1 J

1 kg m² s⁻²

1 A

10⁻¹⁰ m

1 eV

1.602 · 10⁻¹⁹ J

1 g cm⁻³

10⁺³ kg m⁻³



(A)

Vol of block

$$\text{mass of block} = \text{density} \times \text{volume of block}$$

$$\text{volume of block} = \frac{\text{mass of block}}{\text{density}}$$

$$V = \frac{1}{\rho}$$

$$V = 1.1 \cdot 10^{-4} \text{ m}^3$$

Now length of wire (L) → Volume of wire (V)

(B)

$$\text{Area of cross section} \times \text{length} = V$$

$$V = \frac{\pi r^2 L}{4}$$

$$L = \frac{V \cdot 4}{\pi r^2}$$

$$L = 1.4 \cdot 10^{-2} \text{ m}$$

$$L = 1.4 \cdot 10^{-2} \text{ m}$$